📄 Operational Concept Description (OCD)

1.0 Introduction

1.1 Background

In modern automotive manufacturing environments, inspection processes are critical for ensuring product quality and compliance with standard operating procedures (SOPs). Currently, inspection performance is largely dependent on human operators, and there is limited visibility into how thoroughly inspections are conducted. This lack of visibility creates challenges in verifying whether all required inspection zones have been observed and whether inspections are performed consistently across operators.

Manual auditing methods are time-consuming, subjective, and difficult to scale. As a result, there is a need for a system that can objectively monitor inspection behavior and provide measurable insights into inspection quality.

The proposed system aims to support inspection auditing by determining where an inspector directs their visual attention during inspection tasks. This enables the generation of analytics such as inspection coverage, missed areas, and behavioral patterns, improving both quality assurance and training processes.

1.2 Assumptions and Constraints

The system is expected to operate under the following assumptions and constraints:

- Inspectors perform tasks in indoor factory environments with varying lighting conditions
- Inspectors are mobile and interact freely with the vehicle during inspection
- The system must not interfere with normal inspection workflow
- Data processing may occur after inspection completion (non-real-time operation acceptable)

- Network availability is assumed for post-session data transfer, but continuous connectivity is not required
- The system must operate within ergonomic and usability constraints for wearable devices

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2.0 Stakeholders and Operational Actors

Primary Actors

- **Inspector (User):**
  Performs inspection tasks while using the system. The inspector interacts with the system passively, without requiring manual input during operation.
- **Quality Engineer / Supervisor:**
  Reviews inspection analytics to evaluate inspection performance, identify missed areas, and ensure compliance with SOPs.

Supporting Actors

- **Data Analyst / System Operator:**
  Accesses processed inspection data, generates reports, and supports performance evaluation.
- **External Systems:**
  Data storage and processing systems that receive inspection data and provide analytics outputs.

3.0 Operational Overview

The system operates as a passive observation tool that records the inspector's visual behavior during inspection tasks.

At the start of an inspection session, the inspector equips the system and proceeds with standard inspection procedures. The system operates in the background, observing the inspector's visual attention without requiring any manual interaction.

During inspection, the system determines where the inspector is looking over time. This information is used to track inspection behavior and ensure that all required areas are visually inspected.

After the inspection session is completed, the recorded data is processed to generate analytics such as:

- Visual attention heatmaps
- Inspection scanpaths
- Coverage of required inspection zones
- Identification of missed areas

These outputs are then made available to relevant stakeholders for review and decision-making.

The system enables objective evaluation of inspection quality without disrupting existing workflows.

3.1 System Context Diagram (Description)

The system operates within a manufacturing inspection environment and interacts with the following external entities:

- Inspector → provides visual behavior during inspection
- Vehicle / Inspection Object → provides inspection context
- Data Processing System → receives inspection data and generates analytics
- Quality Engineer → receives outputs for evaluation

The system boundary includes all functionality required to observe inspection behavior and produce analytics, while external systems handle broader operational processes.

4.0 Physical and Environmental Context

The system operates in an indoor industrial environment. These environments may include:

- Variable lighting conditions (bright, dim, reflective surfaces)
- Movement of personnel and equipment
- Noise and distractions
- Extended inspection durations

The system is expected to:

- Operate normally under typical indoor conditions
- Maintain functionality under moderate lighting variation
- Tolerate motion and positional changes during inspection

5.0 Operational Scenarios

Scenario 1 — Nominal Operation

An inspector begins a routine inspection of a vehicle. The system is activated and passively observes the inspector's visual attention throughout the process.

The inspector moves around the vehicle, visually examining different components. The system continuously tracks the inspector's gaze behavior and records inspection data.

After completing the inspection, the session ends. The system processes the recorded data and generates analytics, including heatmaps and inspection coverage metrics. The quality engineer reviews the results to verify inspection completeness.

Scenario 2 — Lighting Variation

During inspection, lighting conditions change due to reflections or environmental variation. The system continues to determine visual attention despite these variations and maintains operational performance.

Scenario 3 — Incomplete Inspection

An inspector unintentionally skips certain inspection areas. The system identifies gaps in visual coverage and flags missed zones in the generated report. This allows supervisors to detect incomplete inspections.

Scenario 4 — Data Transfer Delay

Following inspection, data transfer is delayed due to temporary network unavailability. The system retains inspection data and completes transfer once connectivity is restored, ensuring no loss of information.

6.0 Verification Methods

System performance will be verified through a combination of:

- Testing to evaluate accuracy of visual attention tracking
- Analysis to confirm correct generation of inspection analytics
- Demonstration to validate usability and operational behavior

Verification activities will ensure that the system meets stakeholder expectations and supports inspection evaluation effectively.

7.0 Risks and Open Issues

Key risks include:

- Variability in environmental conditions affecting system performance
- Potential discomfort or improper use by inspectors
- Data synchronization challenges affecting accuracy
- Dependence on network availability for data transfer

These risks will be addressed through design decisions, operational procedures, and system validation.

8.0 Candidate System Concepts

Three candidate system concepts were considered:

- **Concept 1:** Local data capture with offline analysis
- **Concept 2:** On-device processing and real-time analytics
- **Concept 3:** Data capture with centralized processing

Each concept differs in how inspection data is processed and how system resources are utilized.

9.0 Trade Study Results

The candidate concepts were evaluated using criteria including:

- Gaze accuracy potential
- Robustness to environment
- Inspector usability
- System reliability

- Research extensibility

The evaluation showed that the centralized processing approach provides the best balance between performance, scalability, and flexibility.

10.0 Selected System Concept

The selected concept is the centralized processing architecture (Concept 3), in which inspection data is collected during operation and processed externally.

This concept was selected because it:

- Supports advanced analytics capabilities
- Reduces operational constraints during inspection
- Provides scalability across multiple inspection sessions
- Aligns with system performance and extensibility requirements

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11.0 Risks

Residual risks include:

- Dependence on external systems for data processing
- Environmental variability affecting performance
- User-related variability in system usage

These risks are considered manageable within the operational context.